INCH-POUND

MIL-PRF-1/1516B
22 May 2007
SUPERSEDING
MIL-E-1/1516A
14 May 1970

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5963

Inactive for new design
after 30 April1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Twin triode, miniature.

Outline - - - 6-2 (EIA).
Base - - - E9-1.
Envelope - - - T6-1/2.
Cathode - - - Coated unipotential.
Base connections:

Pin No. - - -	1	2	3	4	5	6	7	8	9
Element - - -	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Ik/k	Rb/b	Rg/g	Pp/a	Pg/g	TE	Alt
Unit:	V	V dc	V dc	v	mA dc	MegΩ	MegΩ	W	W	°C	ft
Maximum:	13.9	250	0	90	20	---	0.5	2.5	0.5	120	(Note 2)
	6.9	---	---	---	---	---	(Note 1)	---	---	---	---
Minimum:	11.3	---	-100	---	---	---	---	---	---	---	---
	5.7	---	---	---	---	---	---	---	---	---	---
		Ebb	Ecc								
		V dc	V dc								
Test conditions:	12.6	150	0	---	---	0.02	0.047	---	---	---	---

GENERAL:

Qualification: Not required.

AMSC N/A

FSC 5960

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TABLE I. Requirements and tests.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 8)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 1						
1266	Total grid current	Eb = 250 V dc; Ecc = -9 V dc; Rg = 0.5 MegΩ; Rb = 0 (See note 4)	0.65	Ic	---	-1.5	µA dc
1306	Transconductance (1)	Eb = 67.5 V dc; Rb/b = 0; Rg/g = 0	0.65	Sm	2,600	3,800	µmhos
1201	Short and discontinuity detection		0.4	---	---	---	---
1301	Heater current		0.65	If	138	162	mA
1336	Heater-cathode leakage	See note 3	0.65	Ihk	---	10	µA dc
1256	Electrode current (1) (anode)	See note 3	0.65	Ib	4.6	5.4	mA dc
1256	Electrode current (2) (anode)	Ecc = -15 V dc (see note 3)	0.65	Ib	---	50	µA dc
	Conformance inspection, part 2						
1211	Insulation of electrodes		2.5	R	100	---	MegΩ
1306	Transconductance (2)	Ef = 11.4 V; Eb = 67.5 V dc; Rg/g = 0; Rb/b = 0 (see note 3)	2.5	$\frac{\Delta Sm}{Ef}$	---	15	%
1316	Amplification factor	Eb = 67.5 V dc; Rb/b = 0; Rg/g = 0 (see note 3)	---	Mu	18	24	---

See footnotes at end of table.

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TABLE I. Requirements and tests - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 8)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 2 - Continued						
1331	Direct-interelectrode capacitance	No shield (note 3)	6.5	Cgp	1.20	1.80	pF
		No shield (note 3)		Cin	1.60	2.20	pF
		No shield, section 1		Cout	0.35	0.65	pF
		No shield, section 2		Cout	0.24	0.46	pF
		No shield		Cg1-g2	---	0.010	pF
1031	Low-frequency vibration	Ebb = 250 V dc; Ec = - 8.5 V dc; Rb = 2,000 ohms; Rg = 0; 10 G; F = 40 Hz (see note 5)	6.5	Ep	---	150	mV ac
1121	Base strain		---	---	---	---	---
2126	Glass strain		4.0	---	---	---	---
1105	Permanence of marking		---	---	---	---	---
	Conformance inspection, part 3						
1506	Heater-cycling life	Ef = 7.5 V; heater in parallel; Ehk = +135 V dc; Ec = Eb = 0; cycle 1 minute "on" 4 minutes "off"	---	---	---	---	---
---	Heater-cycling life-test end point	Heater-cathode leakage	---	Ihk	---	20	μA dc
1521	Survival-rate life	Eb = 250 V dc; Ec = - 9 V dc; Ehk = +135 V dc; Rg/g = 0.5 MegΩ; Rb/b = 0	---	---	---	---	---

See footnotes at end of table.

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TABLE I. Requirements and tests - Continued.

MIL-STD-1311	Requirement or test	Conditions	Acceptance level (see note 13)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3 - Continued						
---	Survival-rate life-test end point	Short and discontinuity detection	---	---	---	---	---
1511	Cathode-interface life	Heaters in parallel (see note 6)	---	---	---	---	---
---	Cathode-interface life-test end point	See note 7	---	Ri	---	25	Ohms
1501	Intermittent life	Eb = 250 V dc; Ec = - 9 V dc; Ehk = +135 V dc; Rg/g = 0.5 MegΩ; Rb/b = 0; TE = 120°C (min)	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Inoperatives	---	---	---	---	---
		Total grid current	---	Ic	---	- 2.0	μA dc
		Heater current	---	If	138	164	mA
		Change in anode current (1) of individual tubes.	---	$\frac{\Delta Ib}{t}$	---	1.0	mA dc
		Change in transconduct-ance (1) of individual tubes	---	$\frac{\Delta Sm}{t}$	---	20	%
		Transconductance (2)	---	$\frac{\Delta Sm}{Ef}$	---	15	%
		Heater-cathode leakage	---	Ihk	---	20	μA dc
		Insulation of electrodes	---	R	50	---	MegΩ
		Average change in transconductance (1)	---	Avg. $\frac{\Delta Sm}{t}$	---	15	%
		Total defectives	---	---	---	---	---

See footnotes at end of table.

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TABLE I. Requirements and tests - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance level (see note 8)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3 - Continued						
---	Intermittent life-test	Inoperatives	---	---	---	---	---
	end points (1,000 hours)	Total grid current	---	Ic	---	- 2.0	μA dc
		Heater current	---	If	138	166	mA dc
		Change in anode current (1) of individual tubes.	---	$\frac{\Delta Ib}{t}$	---	1.0	mA dc
		Change in transconductance (1) of individual tubes	---	$\frac{\Delta Sm}{t}$	---	25	%
		Transconductance (2)	---	$\frac{\Delta Sm}{Ef}$	---	15	%
		Heater-cathode leakage	---	Ihk	---	20	μA dc
		Insulation of electrodes	---	R	25	---	MegΩ
		Total defectives	---	---	---	---	---

NOTES:

1. This value is for operation under fixed-bias conditions; with cathode bias, Rg/g may be 1.0 megohm, maximum.
2. Altitude test not required as operating voltage is less than 280 volts. Requirements covering altitude versus voltage breakdown relationship are specified in the basic document.
3. Test each unit separately.
4. This test to be performed at the conclusion of the holding period.
5. Tie 1k to 2k; 1g to 2g; and 1a to 2a.
6. The life-test sample shall consist of 20 tubes, and no tube failure shall be permitted. In the event of rejection of the first sample due to failure of 1 tube, a second sample of 20 tubes shall be selected from the lot. Acceptance shall be based upon zero defects. A life-test defect is defined as a failure to meet the life-test end-point limits as specified on the tube specification sheet. The life-test sample shall be read at 500 (+48, -24) hours.
7. Ef = 5.7 V (heaters in parallel), Eb = 100 V dc, adjust Ec to obtain an anode current of 1.5 milliamperes.
8. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

NOTE: Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-021)

Review activities:
Army - AR MI
Navy - AS, CG, MC, OS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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